

Assignment 10: Data Scraping

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on data scraping.

Directions

1. Rename this file `<FirstLast>_A10_DataScraping.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure your code is tidy; use line breaks to ensure your code fits in the knitted output.
5. Be sure to **answer the questions** in this assignment document.
6. When you have completed the assignment, **Knit** the text and code into a single PDF file.

Set up

1. Set up your session:
 - Load the packages `tidyverse`, `rvest`, and any others you end up using.
 - Check your working directory

```
#1
library(tidyverse)
library(rvest)

getwd()
```

```
## [1] "/home/guest/EDE_Fall2025"
```

2. We will be scraping data from the NC DEQs Local Water Supply Planning website, specifically the Durham’s 2024 Municipal Local Water Supply Plan (LWSP):
 - Navigate to <https://www.ncwater.org/WUDC/app/LWSP/search.php>
 - Scroll down and select the LWSP link next to Durham Municipality.
 - Note the web address: <https://www.ncwater.org/WUDC/app/LWSP/report.php?pwsid=03-32-010&year=2024>

Indicate this website as the as the URL to be scraped. (In other words, read the contents into an `rvest` webpage object.)

#2

```
website <- read_html("https://www.ncwater.org/WUDC/app/LWSP/report.php?pwsid=03-32-010&year=2024")
```

3. The data we want to collect are listed below:

- From the “1. System Information” section:
- Water system name
- PWSID
- Ownership
- From the “3. Water Supply Sources” section:
- Maximum Day Use (MGD) - for each month

In the code chunk below scrape these values, assigning them to four separate variables.

HINT: The first value should be “Durham”, the second “03-32-010”, the third “Municipality”, and the last should be a vector of 12 numeric values (represented as strings)“.

#3

```
Water_system <-website %>% html_nodes ("div+ table tr:nth-child(1) td:nth-child(2)") %>% html_text()
PWSID <-website %>% html_nodes ("td tr:nth-child(1) td:nth-child(5)") %>% html_text()
Ownership <-website %>% html_nodes ("div+ table tr:nth-child(2) td:nth-child(4)") %>% html_text()
MGD_months <- website %>% html_nodes ('th~ td+ td , th~ td+ td') %>% html_text()
```

4. Convert your scraped data into a dataframe. This dataframe should have a column for each of the 4 variables scraped and a row for the month corresponding to the withdrawal data. Also add a Date column that includes your month and year in data format. (Feel free to add a Year column too, if you wish.)

TIP: Use `rep()` to repeat a value when creating a dataframe.

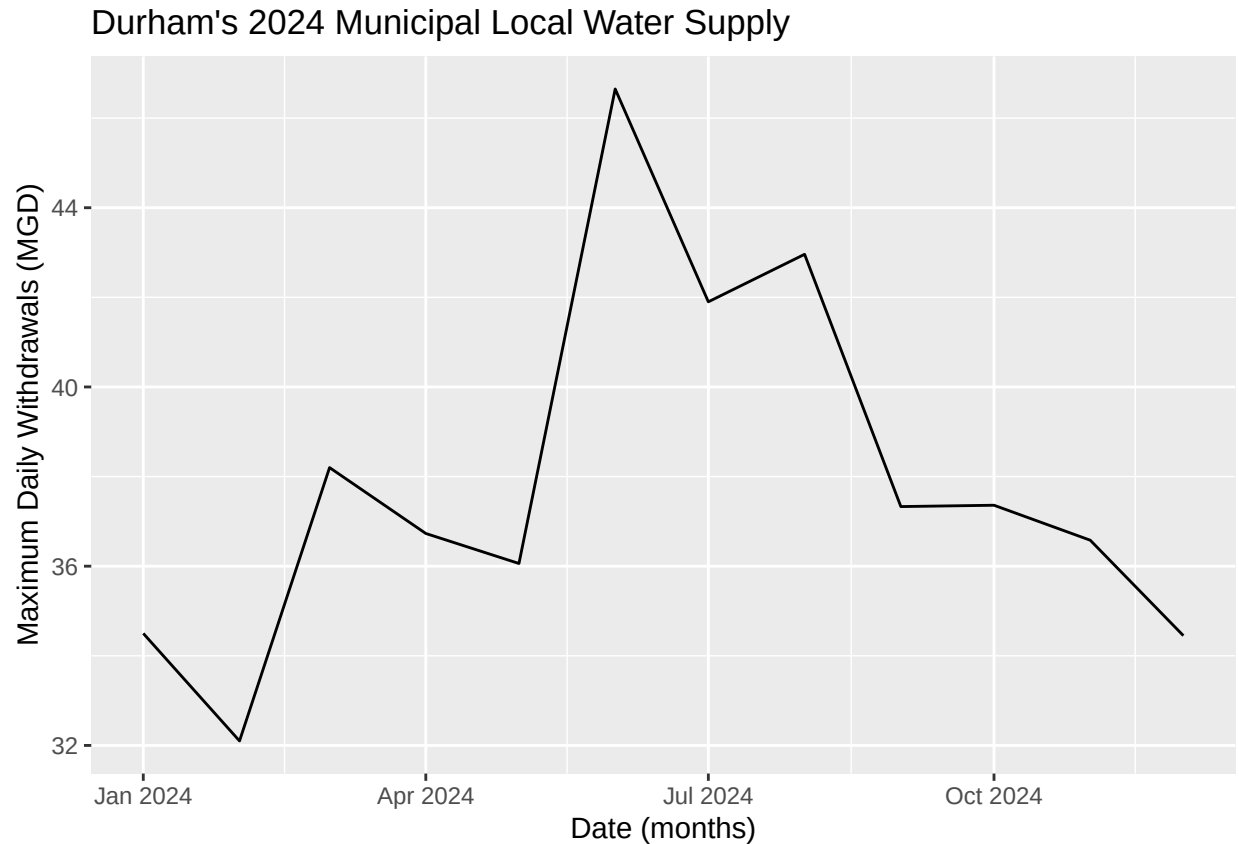
NOTE: It’s likely you won’t be able to scrape the monthly withdrawal data in chronological order. You can overcome this by creating a month column manually assigning values in the order the data are scraped: “Jan”, “May”, “Sept”, “Feb”, etc. . .

5. Create a line plot of the maximum daily withdrawals across the months for 2024, making sure, the months are presented in proper sequence.

#4

```
df_deq <- data.frame("Month" = c(1,5,9,2,6,10,3,7,11,4,8,12),
                     "Year" = rep(2024,12),
                     "Maximum_Day_Use" = as.numeric(MGD_months)) %>%
mutate(Water_System_Name = !!Water_system,
       PWSID = !!PWSID,
       Ownership = !!Ownership,
       Date = my(paste(Month,"-",Year)))
```

```
#5
ggplot(df_deq,aes(x=Date,y=Maximum_Day_Use)) +
  geom_line() +
  labs(x = "Date (months)",
       y = "Maximum Daily Withdrawals (MGD)",
       title = "Durham's 2024 Municipal Local Water Supply")
```



6. Note that the PWSID and the year appear in the web address for the page we scraped. Construct a function with two inputs - “PWSID” and “year” - that:

- Creates a URL pointing to the LWSP for that PWSID for the given year
- Creates a website object and scrapes the data from that object (just as you did above)
- Constructs a dataframe from the scraped data, mostly as you did above, but includes the PWSID and year provided as function inputs in the dataframe.
- Returns the dataframe as the function’s output

```
#6.
scrape.it <- function(the_year, the_pwsid){

  base_url <- "https://www.ncwater.org/WUDC/app/LWSP/report.php"
  scrape_website <- read_html(paste0(base_url, '?pwsid=', the_pwsid, '&year=', the_year))

  Water_system_tag <- "div+ table tr:nth-child(1) td:nth-child(2)"
```

```

PWSID_tag <- "td tr:nth-child(1) td:nth-child(5)"
Ownership_tag <- "div+ table tr:nth-child(2) td:nth-child(4)"
MGD_months_tag <- "th~ td+ td , th~ td+ td"

the_water_system <- scrape_website %>% html_nodes(Water_system_tag) %>% html_text()
the_pwsid_name <- scrape_website %>% html_nodes(PWSID_tag) %>% html_text()
the_ownership_name <- scrape_website %>% html_nodes(Ownership_tag) %>% html_text()
the_mgd_value <- scrape_website %>% html_nodes(MGD_months_tag) %>% html_text()

df_scraped <- data.frame("Month" = c(1,5,9,2,6,10,3,7,11,4,8,12),
                        "Year" = rep(the_year,12),
                        "Maximum_Day_Use" = as.numeric(the_mgd_value)) %>%
mutate(Water_System_Name = !!the_water_system,
       PWSID = !!the_pwsid_name,
       Ownership = !!the_ownership_name,
       Date = my(paste(Month,"-",Year)))

Sys.sleep(1)

return(df_scraped)
}

```

7. Use the function above to extract and plot max daily withdrawals for Durham (PWSID='03-32-010') for each month in 2020. Then show the structure of the dataframe with either the `str()` or the `glimpse()` function.

```

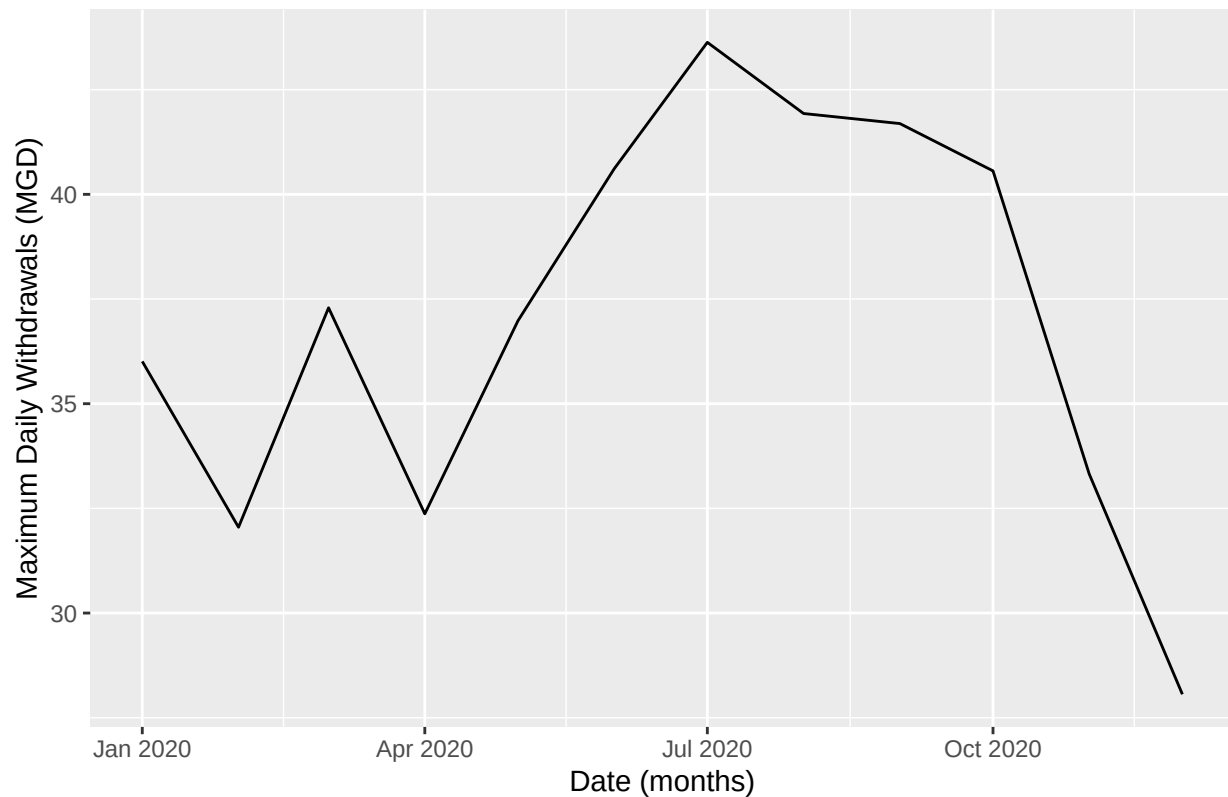
#7
durham_2020 <- scrape.it(2020,"03-32-010")
glimpse(durham_2020)

## Rows: 12
## Columns: 7
## $ Month          <dbl> 1, 5, 9, 2, 6, 10, 3, 7, 11, 4, 8, 12
## $ Year           <dbl> 2020, 2020, 2020, 2020, 2020, 2020, 2020, 2020, 2020, 2020, 2020~
## $ Maximum_Day_Use <dbl> 36.01, 36.98, 41.69, 32.05, 40.61, 40.56, 37.29, 43.~
## $ Water_System_Name <chr> "Durham", "Durham", "Durham", "Durham", "Durham", "Durham", "D~
## $ PWSID          <chr> "03-32-010", "03-32-010", "03-32-010", "03-32-010", ~
## $ Ownership       <chr> "Municipality", "Municipality", "Municipality", "Mun~
## $ Date            <date> 2020-01-01, 2020-05-01, 2020-09-01, 2020-02-01, 2020~

ggplot(durham_2020)+
  geom_line(aes(x = Date, y = Maximum_Day_Use)) +
  labs(x = "Date (months)",
       y = "Maximum Daily Withdrawals (MGD)",
       title = "Durham's 2020 Municipal Local Water Supply")

```

Durham's 2020 Municipal Local Water Supply



- Use the function above to extract data for Cary (PWSID = '03-92-010') in 2020. Combine this data with the Durham data collected above and create a plot that compares Cary's to Durham's water withdrawals.

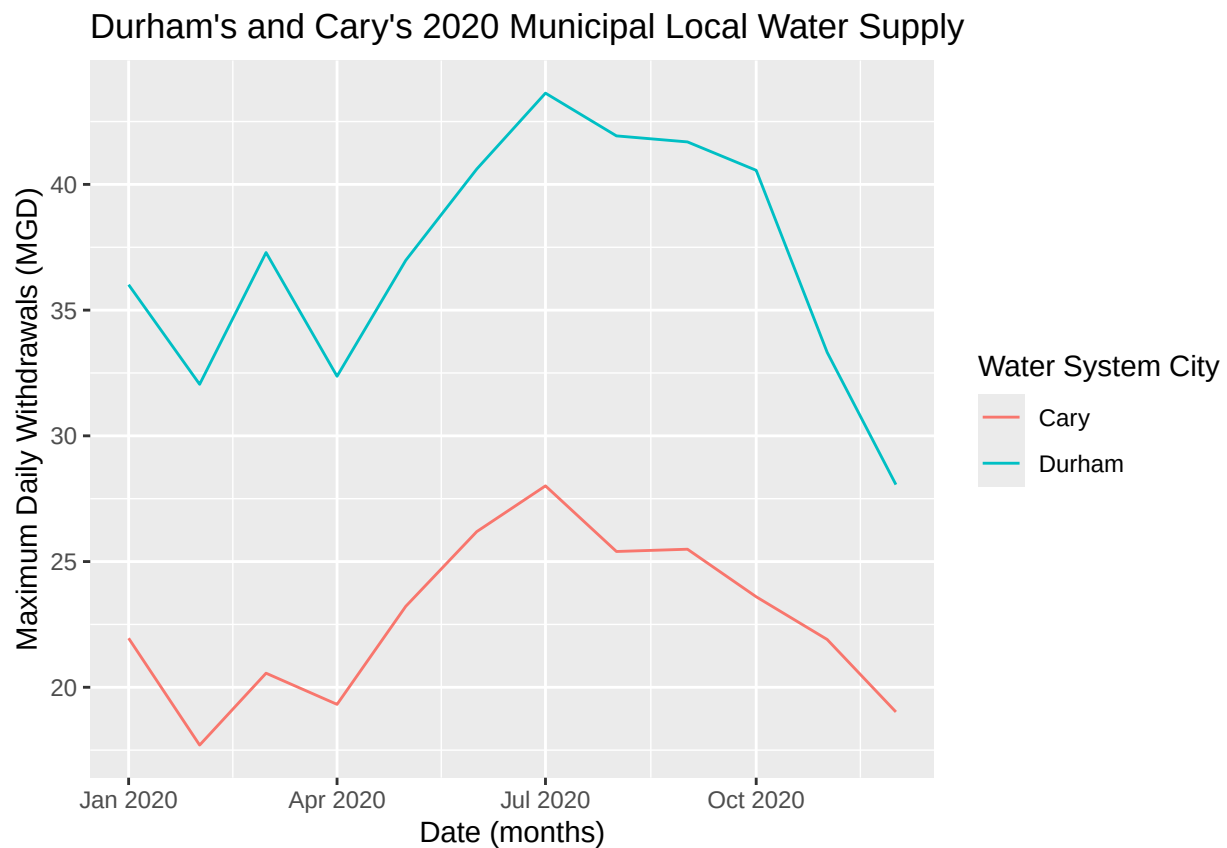
```
#8
cary_2020 <- scrape.it(2020,"03-92-020")
glimpse(cary_2020)
```

```
## Rows: 12
## Columns: 7
## $ Month          <dbl> 1, 5, 9, 2, 6, 10, 3, 7, 11, 4, 8, 12
## $ Year           <dbl> 2020, 2020, 2020, 2020, 2020, 2020, 2020, 2020, 2020, 2020, 2020, 2020~
## $ Maximum_Day_Use <dbl> 21.95, 23.22, 25.49, 17.70, 26.19, 23.60, 20.56, 28.~
## $ Water_System_Name <chr> "Cary", "Cary", "Cary", "Cary", "Cary", "Cary", "Car~
## $ PWSID           <chr> "03-92-020", "03-92-020", "03-92-020", "03-92-020", ~
## $ Ownership       <chr> "Municipality", "Municipality", "Municipality", "Mun~
## $ Date            <date> 2020-01-01, 2020-05-01, 2020-09-01, 2020-02-01, 2020~
```

```
combined_df <- rbind(durham_2020, cary_2020)
```

```
ggplot(data = combined_df)+
  geom_line(aes(x=Date, y= Maximum_Day_Use, color = Water_System_Name)) +
  labs(x = "Date (months)",
       y = "Maximum Daily Withdrawals (MGD)",
```

```
title = "Durham's and Cary's 2020 Municipal Local Water Supply",
color = "Water System City")
```



9. Use the code & function you created above to plot Cary's max daily withdrawal by months for the years 2018 thru 2023. Add a smoothed line to the plot (method = 'loess').

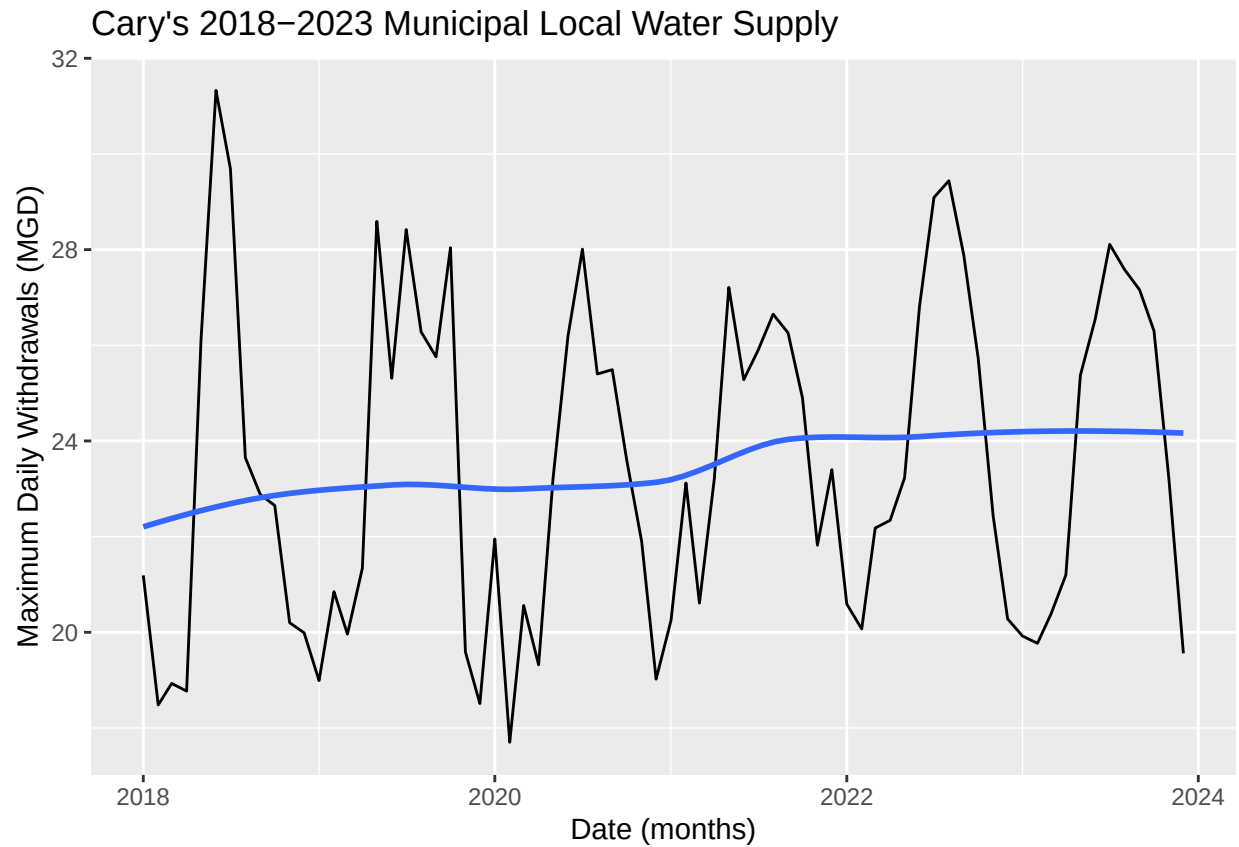
TIP: See Section 3.2 in the "10_Data_Scraping.Rmd" where we apply "map2()" to iteratively run a function over two inputs. Pipe the output of the map2() function to bind_rows() to combine the dataframes into a single one, and use that to construct your plot.

```
#9
the_years = rep(2018:2023)
cary_pwsid = "03-92-020"

the_dfs <- map(the_years, scrape.it, the_pwsid=cary_pwsid)
the_df <- bind_rows(the_dfs)

ggplot(the_df, aes(x=Date, y=Maximum_Day_Use)) +
  geom_line() +
  geom_smooth(method="loess", se=FALSE) +
  labs(x = "Date (months)",
       y = "Maximum Daily Withdrawals (MGD)",
       title = "Cary's 2018-2023 Municipal Local Water Supply")
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



Question: Just by looking at the plot (i.e. not running statistics), does Cary have a trend in water usage over time? > Answer: Just by looking at the plot, Cary has a positive upwards trend in water usage over time as shown by the smoothed line on the graph. >